

MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 20 – 22 | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CLO 5, CLO 8) 2. Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) 3. Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a. (CLO 3, CLO 8) 4. Interpret a p-value in context and relate its size to the strength of evidence against H_0. (CLO 4, CLO 8) 5. Compare the p-value to α, decide whether to reject H_0, write a two-step conclusion in context, and avoid saying “accept H_0”. (CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes “What is a p-value?” video) StatKey – Randomization Hypothesis Tests	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed Participation: P-values and Conclusions – resveratrol examples ($p = 0.04, 0.12, 0.0003$): strength of evidence, decision at α, conclusion in context	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3	WileyPLUS StatKey Lecture Slides (course website) Zoom (live class) Assignment Tool in Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 8** Perform hypothesis testing using statistical methods.